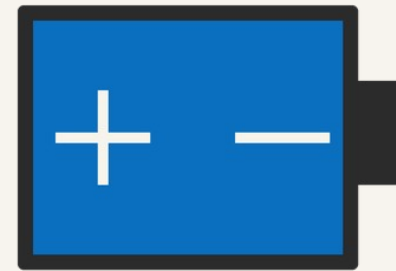


EMF & Internal Resistance

Why a real battery never quite delivers the voltage printed on the label.

Higher Physics



Starter

2007 Higher Physics

Scan the code and answer these:

Q9

Q28

Section A · resistors, wavelength
Show your working, not just the letter.



**Scan for the paper
Marking scheme**

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PART 1

EMF and internal resistance

Every real power supply has a little resistance of its own — and that changes everything downstream of it.

By the end of this part...

- 1 I can define e.m.f., internal resistance, lost volts and terminal p.d.
- 2 I can state the meaning of ideal supply, short circuit and open circuit

The terms

- E.m.f. (E): the total energy the supply gives to each coulomb of charge — measured with NO current flowing.
- Internal resistance (r): the resistance inside the supply itself.
- Lost volts: the energy per coulomb used up pushing charge through r .
- Terminal p.d. (V): what's left for the external circuit — e.m.f. minus lost volts.

Ideal, short circuit, open circuit

- An ideal supply has zero internal resistance — terminal p.d. always equals e.m.f.
- A short circuit has zero external resistance — current is at its maximum, E/r .
- An open circuit has no complete path — no current flows, so terminal p.d. equals e.m.f.

Your turn – the terms

SHOW ME BOARDS

- 1 State what e.m.f. means.
- 2 State what happens to terminal p.d. when the external circuit is open.
- 3 State the internal resistance of an ideal supply.

TRY THIS – Answers

- 1 The energy given to each coulomb of charge by the supply.
- 2 It equals the e.m.f. (no current, so no lost volts).
- 3 Zero.

Starter

2013 Revised Higher Physics

Scan the code and answer these:

Q3

Q17

Section A · uncertainty, circuit reading
Show your working, not just the letter.



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PART 2

The e.m.f. equation

One relationship connects everything a real supply does — e.m.f., lost volts and terminal p.d. all in one line.

By the end of this part...

- 1 I can use $E = V + Ir$ to solve problems involving e.m.f., lost volts and terminal p.d.

$$E = V + Ir$$

$$E = V + Ir$$

E

e.m.f. (V)

V

terminal
p.d. (V)

I

current (A)

r

resistance, r
(Ω)

Ir is the lost volts — rearrange to $V = E - Ir$ when that's what the question asks for.

Worked example – terminal p.d.

A battery of e.m.f. 12 V and internal resistance 0.50 Ω delivers a current of 2.0 A. Find the terminal p.d.

$$E = 12 \text{ V}$$

$$r = 0.50 \text{ } \Omega$$

$$I = 2.0 \text{ A}$$

$$V = ?$$

$$V = E - Ir$$

$$V = 12 - (2.0 \times 0.50)$$

$$\underline{V = 11 \text{ V}}$$

Watch Out — $E = V + Ir$

THE MISTAKE

- Using the external resistance R instead of the current I in Ir
- Assuming terminal p.d. always equals e.m.f.

THE CORRECT ANSWER

- Ir uses the CURRENT, not a resistance value on its own.
- $V = E$ only when $I = 0$ or $r = 0$.

Your turn — $E = V + Ir$

SHOW ME BOARDS

- 1 A cell has e.m.f. 6.0 V and internal resistance $1.0\ \Omega$. Find the terminal p.d. when it delivers 1.5 A .
- 2 State the lost volts in the situation above.
- 3 State what E equals when no current flows.

TRY THIS – Answers

- 1 4.5 V.
- 2 1.5 V.
- 3 The terminal p.d. (they are equal).

Starter

2012 Higher Physics

Scan the code and answer these:



Section A · network resistance

Show your working, not just the letter.



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PART 3

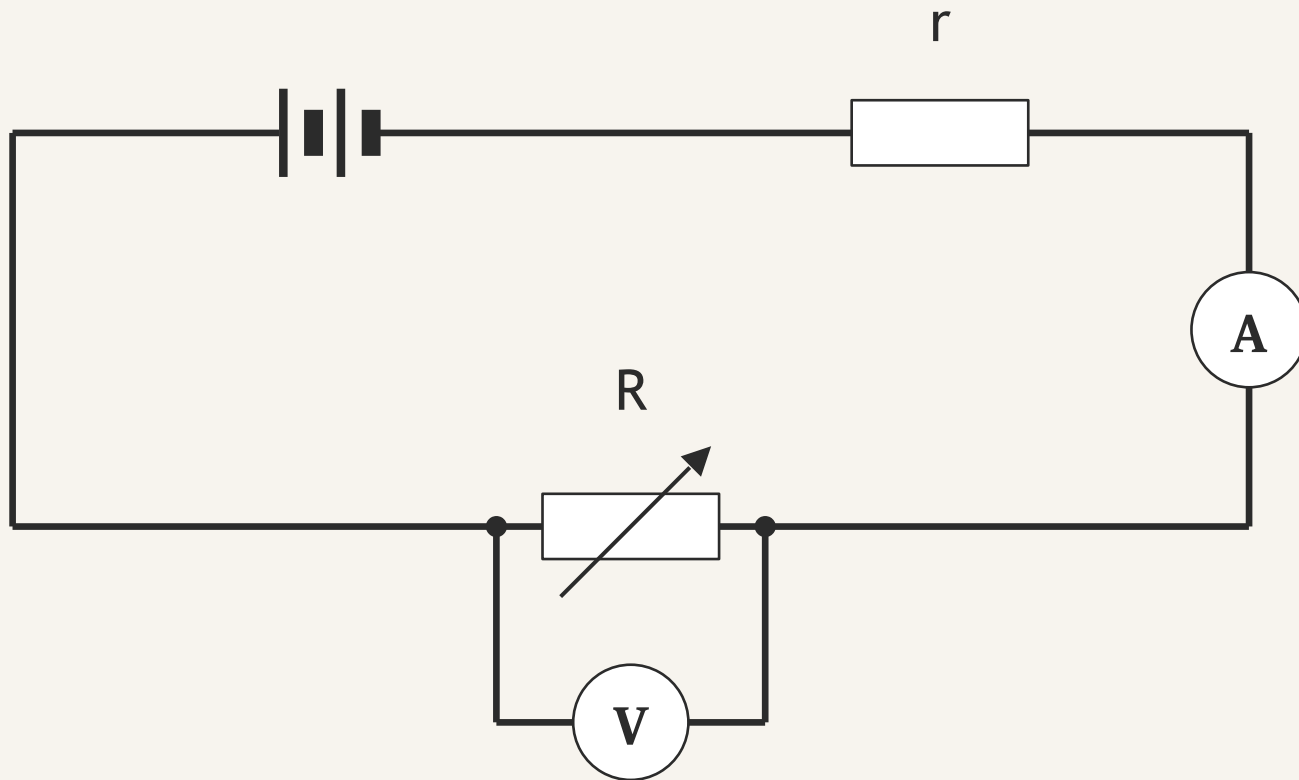
Measuring EMF and r

The named experiment: change the load, read the meters, and both E and r fall out of the same circuit.

By the end of this part...

- 1 I can describe an experiment to measure the e.m.f. and internal resistance of a cell

The circuit



The battery and its internal resistance r are treated as one unit — r cannot be adjusted directly.
 R is a variable resistor: the load the experiment changes.



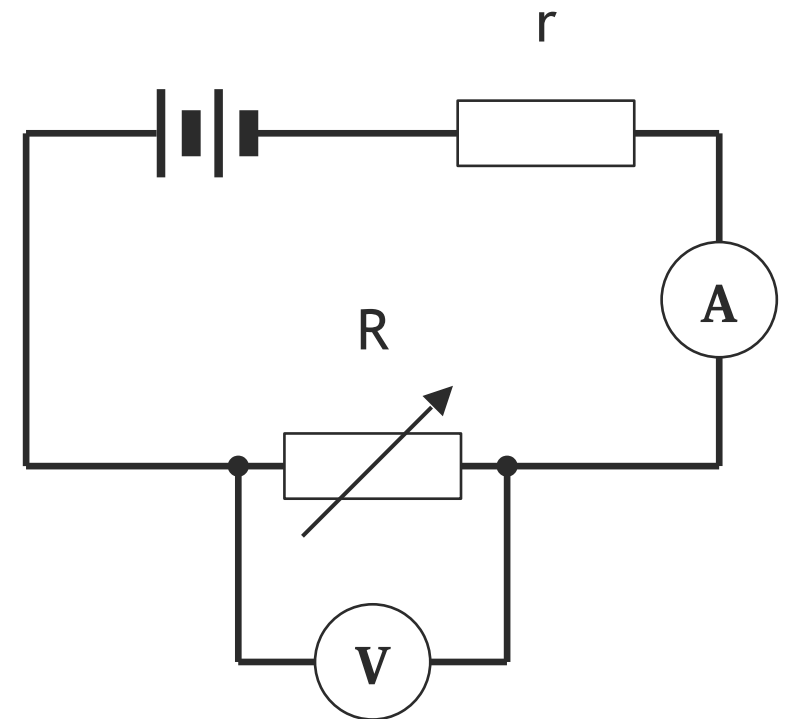
Measuring e.m.f. and internal resistance

Set up a cell, a variable resistor, an ammeter and a voltmeter.

Vary the resistance and record several pairs of V and I .

Plot V (y-axis) against I (x-axis) – a straight line.

y-intercept = E ; gradient = $-r$.



Your turn – the experiment

SHOW ME BOARDS

- 1 State the two meters used in this experiment, and how each is connected.
- 2 State what is varied during the experiment.
- 3 State what quantity is plotted on each axis of the resulting graph.

TRY THIS – Answers

- 1 An ammeter in series, and a voltmeter across the cell's terminals.
- 2 The variable resistor (the external load).
- 3 V (terminal p.d.) on the y-axis, I (current) on the x-axis.

Starter

2014 Revised Higher Physics

Scan the code and answer these:

Q16

Q17

Q20

Section A · resistance, uncertainty
Show your working, not just the letter.



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PART 4

Finding EMF and r from a graph

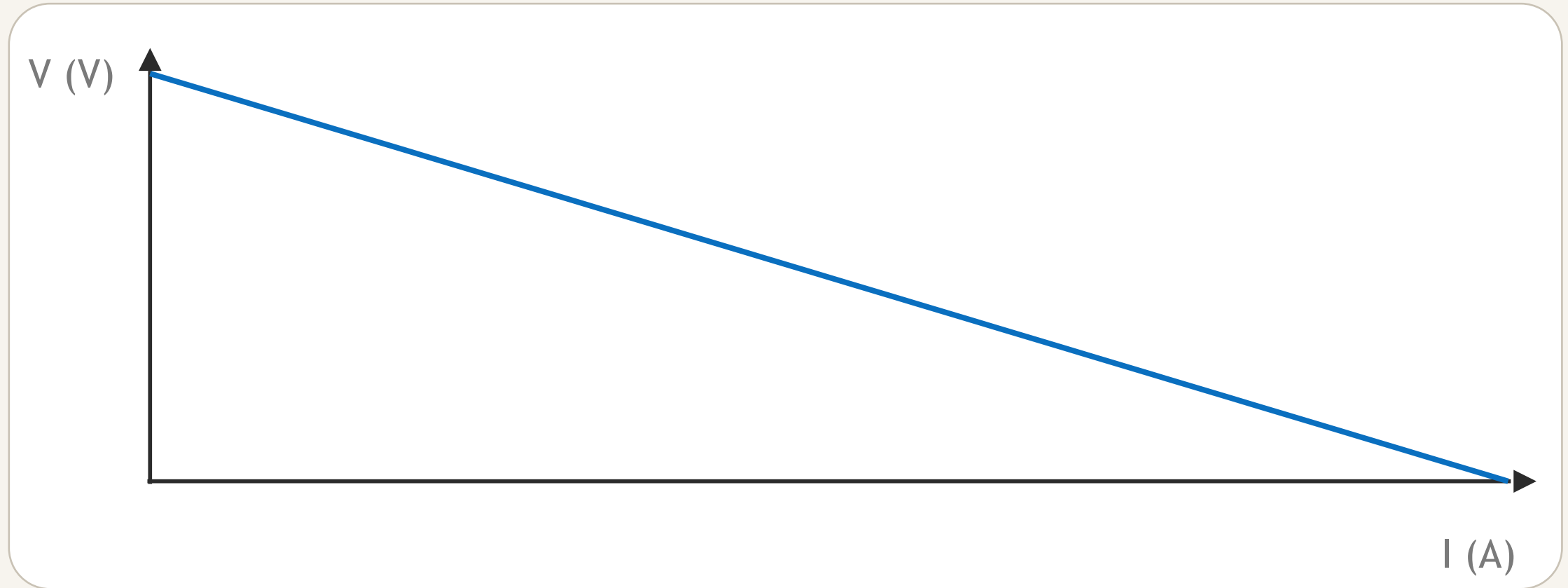
The straight line the experiment produces hides both e.m.f. and internal resistance in plain sight.

By the end of this part...

1

I can determine e.m.f., internal resistance and short-circuit current from a V-I graph

Reading the V-I graph



What each feature means

- The y-intercept ($I = 0$) is the e.m.f., E — no current, so no lost volts.
- The gradient is $-r$ — steeper means MORE internal resistance, not less.
- The x-intercept is the short-circuit current, $I_{sc} = E/r$.

Worked example – reading E and r from a graph

A battery's V - I graph is a straight line from $(0, 6.0 \text{ V})$ to $(12 \text{ A}, 0 \text{ V})$. Find the e.m.f., internal resistance and I_{sc} .

$$V\text{-intercept} = 6.0 \text{ V}$$

$$I\text{-intercept} = 12 \text{ A}$$

$$E = y\text{-intercept} = 6.0 \text{ V}$$

$$r = E/I_{sc}$$

$$r = 6.0 / 12$$

$$r = 0.50 \, \Omega$$

$$\underline{I_{sc} = x\text{-intercept} = 12 \text{ A}}$$

Watch Out – gradient sign

THE MISTAKE

- Reading the gradient as $+r$ instead of $-r$
- Assuming the y-intercept is terminal p.d., not e.m.f.

THE CORRECT ANSWER

- The line slopes down – the gradient is $-r$, not $+r$.
- At $I = 0$ there are no lost volts, so $V = E$.

Your turn – reading the graph

SHOW ME BOARDS

- 1 A V-I graph runs from (0, 9.0 V) to (3.0 A, 0 V). Find the e.m.f.
- 2 For the same graph, find the internal resistance.
- 3 State what the x-intercept of a V-I graph represents.

TRY THIS – Answers

- 1 9.0 V.
- 2 3.0 Ω .
- 3 The short-circuit current.

Starter

2014 Higher Physics

Scan the code and answer these:

Q8

Section A · resistor network

Show your working, not just the letter.



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PART 5

Short circuit and open circuit

The two extremes of the same circuit — no resistance at all, or no path at all.

By the end of this part...

1

I can solve problems involving short-circuit and open-circuit conditions

Worked example – short-circuit current

A battery has e.m.f. 10.0 V and internal resistance $2.0\ \Omega$.
Find the short-circuit current.

$$E = 10.0\text{ V}$$

$$r = 2.0\ \Omega$$

$$R = 0\ \Omega \text{ (short circuit)}$$

$$I = E/r$$

$$I = 10.0 / 2.0$$

$$\underline{I = 5.0\text{ A}}$$

Checkpoint

- 1 Define e.m.f. and internal resistance.
- 2 State the internal resistance of an ideal supply.
- 3 Use $E = V + Ir$ to find lost volts, given E , V and I .
- 4 Describe the experiment used to measure e.m.f. and internal resistance.
- 5 State what the gradient and intercepts of a V - I graph represent.

Exam Practice

2025 Q14 – increasing the current in the experimental setup.

2024 Q14 – E and r from a V - I graph, and short-circuit current.

2023 Q12 – terminal p.d., internal resistance and power.

Past papers, Section A

LOOKING AHEAD

Next: Monitoring & Measuring AC

A supply that never stops changing — and the meters built to keep up with it.

Plenary

- 1 Define e.m.f., internal resistance, lost volts and terminal p.d.
- 2 State $E = V + Ir$.
- 3 Describe the named experiment for measuring E and r .
- 4 State what the gradient and both intercepts of a V - I graph represent.
- 5 State the short-circuit current formula.

- 1 Explain why a car will not start with a battery that has a very high internal resistance, even if its e.m.f. reads normally.
- 2 Explain why an ammeter should have very low resistance and a voltmeter very high resistance in this experiment.

- 1 Derive the short-circuit current formula $I_{sc} = E/r$ from $E = V + Ir$.
- 2 Two identical cells are connected in series. Explain what happens to the combined e.m.f. and internal resistance.

Extension — EXTRA

EXTRA PRACTICE

- 1 Research why a car battery has a much lower internal resistance than a small torch battery.